

숙제1 결과 => 70점 중에 68점 획득

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①  $\vec{r} = \vec{R} + \delta \vec{r} = (R+x)\vec{i} + y\vec{j} + z\vec{k}$ ,  $\vec{w} = w\vec{k}$ .

② a. 속도와 가속도를 증명하라.

$\vec{v} = \langle \dot{x} - w y, \dot{y} + w(R+x), \dot{z} \rangle$  (ijk)  
 $\vec{v} = \langle \dot{x} - 2w y - w^2(R+x), \dot{y} + 2w\dot{x} - w^2 y, \dot{z} \rangle$  (ijk)

LVLH xyz 프레임. 현재  $\vec{r}$ 가 관성계가 아닌 회전계라서  $\left| \frac{d\vec{r}}{dt} \right|_i = \left| \frac{d\vec{r}}{dt} \right|_r + w \times \vec{r}$  을 쓰자.

$\vec{v} = \left| \frac{d\vec{r}}{dt} \right|_i = \langle \dot{x} - w y, \dot{y} + w(R+x), \dot{z} \rangle + \langle 0, 1, w \rangle \times \langle x - w y, \dot{y} + w(R+x), \dot{z} \rangle = \langle \dot{x} - 2w y - w^2 R - w^2 x, \dot{y} + 2w\dot{x} - w^2 y, \dot{z} \rangle$  (7.1)  
 $\vec{v} = \left| \frac{d\vec{r}}{dt} \right|_r = \langle \dot{x}, \dot{y}, \dot{z} \rangle + \langle 0, 0, w \rangle \times \langle R+x, y, z \rangle = \langle \dot{x}, \dot{y}, \dot{z} \rangle + \langle -w y, w(R+x), 0 \rangle = \langle \dot{x} - w y, \dot{y} + w(R+x), \dot{z} \rangle$

③ b.  $\vec{v} = -\frac{\mu}{r^3} \vec{r} + \vec{v}_0$  을 써서 ijk계의 운동방정식을 구하라.  $\mu = 3.98 \times 10^{14} \text{ m}^3/\text{s}^2$ ,  $\vec{v}_0 = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$  = 초기속도,  $r = \sqrt{(R+x)^2 + y^2 + z^2} = r_0/\mu$  가리.

$\ddot{\vec{r}} = -\frac{\mu}{r^3} \langle R+x, y, z \rangle + \langle v_x, v_y, v_z \rangle = \langle -\frac{\mu}{r^3} (R+x) + v_x, -\frac{\mu}{r^3} y + v_y, -\frac{\mu}{r^3} z + v_z \rangle$   
 $\ddot{x} - 2w\dot{y} - w^2(R+x) = -\frac{\mu}{r^3} (R+x) + v_x$   
 $\ddot{y} + 2w\dot{x} - w^2 y = -\frac{\mu}{r^3} y + v_y$   
 $\ddot{z} = -\frac{\mu}{r^3} z + v_z$

④ c. b.식으로부터 R과 w를 나누어 비차원화하라. CANONICAL units.  $r, \dot{r}, \ddot{r}$  을 R로 나누기.  $w$  는  $1/w$ 로 나누기.

원래식)  $\ddot{x} - 2w\dot{y} - w^2(R+x) = -\frac{\mu}{r^3} (R+x) + v_x$   
 새조율식)  $\ddot{x} - 2\dot{y} - (1+x) = -\frac{\mu}{r^3} (1+x) + \bar{v}_x$   
 $\ddot{y} + 2\dot{x} - y = -\frac{\mu}{r^3} y + \bar{v}_y$   
 $\ddot{z} = -\frac{\mu}{r^3} z + \bar{v}_z$

$r = \sqrt{(1+x)^2 + y^2 + z^2}$   
 $\mu = 1$  : CANONICAL 단위

⑤ CW 식을 얻기 위해서, 원계로 대치하여, CANONICAL 단위로, 운동방정식을 구하라. (선형화식)

$\bar{x}, \bar{y}, \bar{z}$  = 원점 으로 설정.  $\bar{x} = 0 + \Delta x$ ,  $\bar{y} = 0 + \Delta y$ ,  $\bar{z} = 0 + \Delta z$ .

$\Delta \ddot{x} - 2\Delta \dot{y} - (1+\Delta x) = -\frac{1+\Delta x}{[(1+\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2]^{1.5}} + \bar{v}_x = -\frac{(1+\Delta x)}{(1+\Delta x)^3} + \bar{v}_x = -(1+\Delta x)^{-2} + \bar{v}_x = -(-2\Delta x) + \bar{v}_x$   
 $\Delta \ddot{y} + 2\Delta \dot{x} - \Delta y = -\frac{\Delta y}{[(1+\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2]^{1.5}} + \bar{v}_y = -\Delta y + \bar{v}_y \rightarrow \Delta \ddot{y} + 2\Delta \dot{x} = \bar{v}_y$   
 $\Delta \ddot{z} = -\frac{\Delta z}{[(1+\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2]^{1.5}} + \bar{v}_z = -\Delta z + \bar{v}_z \rightarrow \Delta \ddot{z} + \Delta z = \bar{v}_z$

CW식 유도완료!  
 (단,  $\Delta$ 와  $\bar{v}$ 만 제거)  
 $\bar{v} \rightarrow F$ 로 대체  
 $r, \dot{r}, \ddot{r}$

② 강의로 5쪽에서  $r, \dot{r}, \ddot{r}$  과 중력, 항력으로 구한

③ 91페이지 케도면에 고정되어 고도 600km에 있다.  $\lambda=0, F=0$ .  
 $r = 600 + 600 = 1200 \text{ km}$ ,  $\dot{r} = \ddot{r} = 0$ .

④ a. CONSTANT ORBITAL VELOCITY=?

2번식)  $\ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2 = -\frac{\mu}{r^2} + \frac{F_r}{m}$   
 $r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda} \sin \lambda = \frac{F_\lambda}{m}$   
 $r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\lambda}^2 \sin \lambda \cos \lambda = \frac{F_\lambda}{m}$

$r\dot{\lambda}^2 = -\frac{400000}{r^2} \rightarrow \dot{\lambda} = \sqrt{\frac{\mu}{r^3}} = \sqrt{\frac{400000}{1200^3}} = 0.00108 \text{ (rad/s)} \Rightarrow V = r\dot{\lambda} = 1200 \times 0.00108$   
 $r\ddot{\lambda} = 0 \rightarrow \ddot{\lambda} = 0, \dot{\lambda} = \text{상수}, \lambda = \text{상수} \cdot t + \text{상수}$

⑤ b. 케도주기=?  $T = \frac{2\pi}{\dot{\lambda}} = \frac{2\pi}{0.00108 \text{ (rad/s)}} = \text{주기} = \frac{0.28 \text{ (rad)}}{0.00108 \text{ (rad/s)}} = 5815 \text{ 초} = 1.615 \text{ 시간}$



⑤ 위성이  $6400+600=7000$  km 높이에 원궤도를 돈다. 갑자기 궤도 속도가 방향대로 크지만 1% 커졌다. ③

선형 섭동역학 문제 4번 맨 아래부터!

$$\vec{v} \Rightarrow 1.01 \vec{v}$$

①  $\vec{r}$  와  $\vec{v}$  = ? 1회전 동안, 외력없음 같은 궤도면

IN-plane

문제 4번은 CW eq를 이용하는 데, 우리 5번은  $F=0$  이라고 제시 했기때, Free orbital motion ~~이~~ 에 해당된다.

$$\begin{bmatrix} x(t) \\ v_x(t) \\ y(t) \\ v_y(t) \end{bmatrix} = \begin{bmatrix} 4-3\cos w_0 t & \frac{\sin w_0 t}{w_0} & 0 & \frac{2(1-\cos w_0 t)}{w_0} \\ 3w_0 \sin w_0 t & \cos w_0 t & 0 & 2\sin w_0 t \\ 6\sin w_0 t - 6w_0 t & \frac{2(1+\cos w_0 t)}{w_0} & 1 & \frac{4\sin w_0 t}{w_0} - 3t \\ 6w_0(1+\cos w_0 t) & -2\sin w_0 t & 0 & -3+4\cos w_0 t \end{bmatrix} \begin{bmatrix} x(0) \\ v_x(0) \\ y(0) \\ v_y(0) \end{bmatrix}$$

$w_0 = 0.00108$  (rad/s) ( $\therefore$  4번에서 구했음)

$\rightarrow$  문제에서 제시

$\rightarrow$  1% 증가  $\rightarrow v_y(0) = \frac{v}{100} = \frac{7000 \cdot 0.00108}{100} = 0.0756$  (km/s)

$$\begin{bmatrix} x(t) \\ v_x(t) \\ y(t) \\ v_y(t) \end{bmatrix} = \begin{bmatrix} \frac{2(1-\cos w_0 t)}{w_0} \\ 2\sin w_0 t \\ \frac{4\sin w_0 t}{w_0} - 3t \\ -3+4\cos w_0 t \end{bmatrix} v_{y0}$$

$x = \Delta r$   
 $y = R \Delta \mathcal{D}$   
 $z = R \Delta \lambda$

$$\begin{aligned} r(t) &= R + \Delta r = R + x = R + \frac{2(1-\cos w_0 t)}{w_0} v_{y0} \\ \mathcal{D}(t) &= \sqrt{\frac{a}{R^3}} t + \Delta \mathcal{D} = w_0 t + \frac{y}{R} = w_0 t + \frac{1}{R} \left( \frac{4\sin w_0 t}{w_0} - 3t \right) v_{y0} \\ \lambda(t) &= 0 + \Delta \lambda = 0 \end{aligned}$$

$$\begin{aligned} \dot{r}(t) &= \Delta \dot{r} = 2\sin w_0 t \cdot v_{y0} \\ \dot{\mathcal{D}}(t) &= w_0 + \Delta \dot{\mathcal{D}} = w_0 + \frac{1}{R} (-3 + 4\cos w_0 t) v_{y0} \\ \dot{\lambda}(t) &= 0 \end{aligned}$$

⑥  $r_p$   $v_a$   $e$   $T$   
 근지점, 원지점, 이심률, 주기 = ?

$$r(t) = R + \frac{2(1-\cos w_0 t)}{w_0} v_{y0} = 7000 + \frac{2-2\cos w_0 t}{0.00108} \cdot 0.0756 \text{ (km)} \Rightarrow \text{각각 근지/원지에 해당한다. 즉, } \cos w_0 t = 1 \text{ 이면 } r_p$$

$$\therefore \cos w_0 t = 1, r(t) = R + 0 = 7000 \text{ km} = r_p = \text{근지점.}$$

$$\therefore \cos w_0 t = -1, r(t) = R + 11 = 7000 + \frac{4}{0.00108} \cdot 0.0756 = 7280 \text{ km} = r_a = \text{원지점.}$$

1.66 시간.

$$e = \frac{r_a - r_p}{r_a + r_p} = \frac{280}{14280} = 0.0196 = 0.02, a = \frac{r_a + r_p}{2} = \frac{14280}{2} = 7140 \text{ km}, T = 2\pi \sqrt{\frac{a^3}{\mu}} = 6.28 \cdot \frac{7140^3}{\sqrt{400,000}} = 5490 \text{ s}$$

④ CW!

$$\begin{pmatrix} \ddot{x} - 2w\dot{y} - 3w^2x = \frac{F_r}{m} \\ \ddot{y} + 2w\dot{x} = \frac{F_{\mathcal{D}}}{m} \\ \ddot{z} + w^2z = \frac{F_{\lambda}}{m} \end{pmatrix}$$

$w = \text{회전각속도}$   
 $F = \text{중력}$   
 $w = \dot{\mathcal{D}} = \sqrt{\frac{\mu}{R^3}} = 0.00108$  (rad/s)  
 공전궤도속도

$x = 1 \cdot \Delta r$   
 $y = R \Delta \mathcal{D}$   
 $z = R \Delta \lambda$

6. 문제3의 선형 미방으로,  $F_r, F_d$ 를 도입하는 것일뿐. controllability를 판단해라. 그리고 결론을 해석해라.

(a) local vertical propulsive  $F_r$  (b) local horizontal propulsive  $F_d$  (c)  $F_r$ 과  $F_d$

state-space form in plane D

$$\begin{bmatrix} \dot{x} \\ \dot{V}_x \\ \dot{y} \\ \dot{V}_y \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 3W_0^2 & 0 & 0 & 2W_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2W_0 & 0 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ V_x \\ y \\ V_y \end{bmatrix}}_X + \underbrace{\begin{bmatrix} 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m} \end{bmatrix}}_B \begin{bmatrix} F_r \\ F_d \end{bmatrix}, \quad \text{rank} [B \ AB \ \dots \ A^{n-1}B] = 4, \text{ completely con.}$$

(a)  $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3W_0^2 & 0 & 0 & 2W_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2W_0 & 0 & 0 \end{bmatrix}$   $B = \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \\ 0 \end{bmatrix}$   $A^2 = \begin{bmatrix} 3W_0^2 & 0 & 0 & 2W_0 \\ 0 & -W_0^2 & 0 & 0 \\ 0 & -2W_0 & 0 & 0 \\ -6W_0^3 & 0 & 0 & -4W_0^2 \end{bmatrix}$   $A^3 = \begin{bmatrix} 0 & -W_0^2 & 0 & 0 \\ -3W_0^4 & 0 & 0 & -2W_0^3 \\ -6W_0^3 & 0 & 0 & -4W_0^2 \\ 0 & 2W_0^3 & 0 & 0 \end{bmatrix}$   $A^4 = \begin{bmatrix} -3W_0^4 & 0 & 0 & -2W_0^3 \\ 0 & W_0^4 & 0 & 0 \\ 0 & 2W_0^3 & 0 & 0 \\ 6W_0^5 & 0 & 0 & 4W_0^4 \end{bmatrix}$

$\text{rank} [B \ AB \ A^2B \ \dots \ A^{n-1}B] = \begin{bmatrix} 0 & \frac{1}{m} & 0 & 0 \\ 0 & 0 & -\frac{W_0^2}{m} & \frac{2W_0}{m} \\ 0 & 0 & -\frac{2W_0}{m} & 0 \\ 0 & 0 & \frac{2W_0^3}{m} & -\frac{4W_0^2}{m} \end{bmatrix} = 3$

$\downarrow$   
 $A^2B$ 가 0이다  
 $A^3B, A^4B \dots$ 도 0이다  
 $\therefore$  completely con 이 4다

(b)  $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3W_0^2 & 0 & 0 & 2W_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2W_0 & 0 & 0 \end{bmatrix}$   $B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m} \end{bmatrix}$   $A^2 = 9I$   $A^3 = 9I$   $A^4 = 9I$

$\text{rank} [B \ AB \ A^2B \ \dots \ A^{n-1}B] = \begin{bmatrix} 0 & 0 & 2W_0/m & 0 \\ 0 & 0 & -2W_0^3/m & -4W_0^2/m \\ 0 & 0 & -4W_0^2/m & 0 \\ 0 & 0 & 4W_0^3/m & 0 \end{bmatrix} = 4 \rightarrow A^{n-1}B = A^3B \therefore$  completely con.

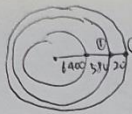
(c)  $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3W_0^2 & 0 & 0 & 2W_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2W_0 & 0 & 0 \end{bmatrix}$   $B = \begin{bmatrix} 0 & 0 \\ 1/m & 0 \\ 0 & 0 \\ 0 & 1/m \end{bmatrix}$   $A^2 = 9I$   $A^3 = 9I$   $A^4 = 9I$

$\text{rank} [B \ AB \ A^2B \ A^3B \ \dots \ A^{n-1}B] = \begin{bmatrix} 0 & 0 & 1/m & 0 \\ 0 & 0 & 0 & 2W_0/m \\ 0 & 0 & 0 & -2W_0^3/m \\ 0 & 0 & 0 & -4W_0^2/m \end{bmatrix} = 4$

$\downarrow$   
 $A^{n-1}B = A^3B \therefore$  completely con.



7.



30 분록 장대부

①  $\Delta V_1, \Delta V_2$ 의 순간속력

② non-impulsive propulsive f.

노트 13쪽의 랑데부가 2번속력 ④ 노트 18쪽의 CO-planar 만개도의 B. ⑤  
14쪽과 동일. (이제는 13쪽에 편지)  
(공속은 14쪽에 편지)

①  $V_1 = 6400 + 580 = 6980, V_2 = 6400 + 600 = 7000 \text{ km/h}, V_1 - V_2 = -20, t_f = 1800 \text{ 초}, W_0 = \sqrt{\frac{M}{Y_2}} = 0.00108 \text{ (N/S)}$

$$\begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix} = \begin{bmatrix} -20 \\ 0 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{\Delta V_0} \begin{bmatrix} x_0 \\ \dot{x}_0^+ \\ y_0 \\ \dot{y}_0^+ \end{bmatrix} = \begin{bmatrix} -20 \\ V_{0x}^+ \\ 0 \\ V_{0y}^+ \end{bmatrix} \xrightarrow{\text{coast phase}} e^{At_f} \begin{bmatrix} x_0 \\ \dot{x}_0^+ \\ y_0 \\ \dot{y}_0^+ \end{bmatrix} = \begin{bmatrix} x_f \\ \dot{x}_f \\ y_f \\ \dot{y}_f \end{bmatrix}$$

③ 
$$\begin{bmatrix} 4-3\cos W_0 t & \frac{\sin W_0 t}{W_0} & 0 & \frac{2(1-\cos W_0 t)}{W_0} \\ 3W_0 \sin W_0 t & \cos W_0 t & 0 & 2\sin W_0 t \\ 6\sin W_0 t - 6W_0 t & \frac{2(1+\cos W_0 t)}{W_0} & 1 & \frac{4\sin W_0 t}{W_0} - 3t \\ W_0(1+\cos W_0 t) & -2\sin W_0 t & 0 & -3+4\cos W_0 t \end{bmatrix} \begin{bmatrix} -20 \\ V_{0x}^+ \\ 0 \\ V_{0y}^+ \end{bmatrix}$$

①  $-80 + 60 \cos W_0 t + \frac{\sin W_0 t}{W_0} \cdot V_{0x}^+ + \frac{2(1-\cos W_0 t)}{W_0} \cdot V_{0y}^+ = 0$

②  $\dot{x}_f^- = -60 W_0 \sin W_0 t + V_{0x}^+ \cos W_0 t + V_{0y}^+ \cdot 2 \sin W_0 t$

③  $0 = -120 \sin W_0 t + 120 W_0 t + V_{0x}^+ \cdot \frac{2(1+\cos W_0 t)}{W_0} + \left( \frac{4 \sin W_0 t}{W_0} - 3t \right) \cdot V_{0y}^+$

④  $\dot{y}_f^- = -120 W_0 (1+\cos W_0 t) - 2 V_{0x}^+ \sin W_0 t + V_{0y}^+ \cdot (-3+4 \cos W_0 t)$

①, ③ 변식오,  $W_0 t = 1.944 \text{ (rad)} = 111.383^\circ, \sin 1.944 = 0.9311, \cos 1.944 = -0.3646$

$80 - 60 \cdot -0.3646 = \frac{0.9311}{0.00108} V_{0x}^+ + \frac{2 \cdot 2 \cdot 0.3646}{0.00108} \cdot V_{0y}^+ \rightarrow 101.876 = 862.129 V_{0x}^+ + 2527.037 V_{0y}^+$

$120 \cdot 0.9311 - 120 \cdot 1.944 = \frac{2 \cdot (-0.3646) - 2}{0.00108} \cdot V_{0x}^+ + \left( \frac{4 \cdot 0.9311}{0.00108} - 5400 \right) \cdot V_{0y}^+ \rightarrow -121.548 = -2527.037 V_{0x}^+ - 1951.482 V_{0y}^+$

연립하여,  $V_{0x}^+ = 0.023054 \text{ km/s}$   
 $V_{0y}^+ = 0.032581 \text{ km/s}$

②, ④ 변식오

$V_{fx}^- = -60 \cdot 0.00108 \cdot 0.9311 + 0.023054 \cdot (-0.3646) + 0.032581 \cdot 2 \cdot 0.9311 = -8.1 \text{ m/s} = V_{fx}^-$

$V_{fy}^- = -120 \cdot 0.00108 \cdot (-1-0.3646) - 2 \cdot 0.9311 \cdot 0.023054 + 0.032581 \cdot (-3-4 \cdot 0.3646) = -11.1 \text{ m/s} = V_{fy}^-$

초기시점의  $\dot{x}_0 = 0, \dot{y}_0 = -1.5 \text{ m/s}, x_0 = 0.032481 \text{ km/s}$  라서  $\Delta V_{0x} = 0.023054 \text{ km/s}$  를,  $\Delta V_{0y} = 0.001 \text{ km/s}$  를 가한다.

중기시점의  $\dot{x}_f = 0, \dot{y}_f = 0$  이라서,  $\Delta V_{fx} = -8.1 \text{ m/s}$  를,  $\Delta V_{fy} = -11.1 \text{ m/s}$  를 가한다.

× ⑥ 노트 14쪽의 piecewise constant 매개변수.  $t_1 = 900 \text{ s}, t_f = 1800 \text{ s}$  A와 B는 노트 9쪽  
노트 18쪽의 CO-planar 만개도의 B 조건.

$$\begin{bmatrix} x_f \\ \dot{x}_f \\ y_f \\ \dot{y}_f \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = e^{At_f} \begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix} + \int_0^{t_f} e^{A(t_f-\tau)} \cdot B \cdot d\tau \begin{bmatrix} F_{r1} \\ F_{p1} \end{bmatrix} + \int_{t_1}^{t_f} e^{A(t_f-\tau)} \cdot B \cdot d\tau \begin{bmatrix} F_{r2} \\ F_{p2} \end{bmatrix}$$

$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3W_0^2 & 0 & 0 & 2W_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2W_0 & 0 & 0 \end{bmatrix}$   
 $B = \begin{bmatrix} 0 & 0 \\ \frac{1}{M} & 0 \\ 0 & 0 \\ 0 & -\frac{1}{M} \end{bmatrix}$

$$\begin{bmatrix} F_{r1} \\ F_{p1} \end{bmatrix} =$$

$$\begin{bmatrix} F_{r2} \\ F_{p2} \end{bmatrix} =$$

수치적분 계산? 노트 14쪽의 W계산을 만났다고 했는데? 음...